

# Project Smart Parking System

## Components:

- Arduino Uno R3
- Arduino 5V
- HC-SR04 Ultrasonic Sensor
- 16x2 LCD Display (parallel, non-I2C)
- Potentiometer (10k $\Omega$ )
- Passive Buzzer
- LEDs
  - Green LED
  - Yellow LED
  - Red LED
- 220 $\Omega$  Resistors
- Breadboard
- Jumper Wires (Male-Male)

## Code

```
#include <LiquidCrystal.h>

LiquidCrystal lcd(2, 3, 4, 5, 6, 7);

// Pins
const int echoPin = 13;
const int trigPin = 12;
const int buzzer = 11;
const int greenLED = 10;
const int yellowLED = 9;
const int redLED = 8;

long duration;
int distance;

void setup() {
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);

  pinMode(buzzer, OUTPUT);
  pinMode(greenLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(redLED, OUTPUT);
}
```

```

    lcd.begin(16, 2);
    Serial.begin(9600);
}

void loop() {

    digitalWrite(trigPin, LOW);
    delayMicroseconds(2);
    digitalWrite(trigPin, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigPin, LOW);

    duration = pulseIn(echoPin, HIGH);
    distance = duration * 0.034 / 2;
    lcd.setCursor(0, 0);
    lcd.print("Distance: ");
    lcd.print(distance);
    lcd.print(" cm  ");

    int beepDelay = map(distance, 10, 40, 50, 500);

    digitalWrite(greenLED, LOW);
    digitalWrite(yellowLED, LOW);
    digitalWrite(redLED, LOW);

    if (distance > 40) {
        lcd.setCursor(0, 1);
        digitalWrite(greenLED, HIGH);
        lcd.print("SAFE      ");
        noTone(buzzer);
    }
    else if (distance > 20) {
        lcd.setCursor(0, 1);
        lcd.print("SLOW DOWN    ");
        digitalWrite(yellowLED, HIGH);

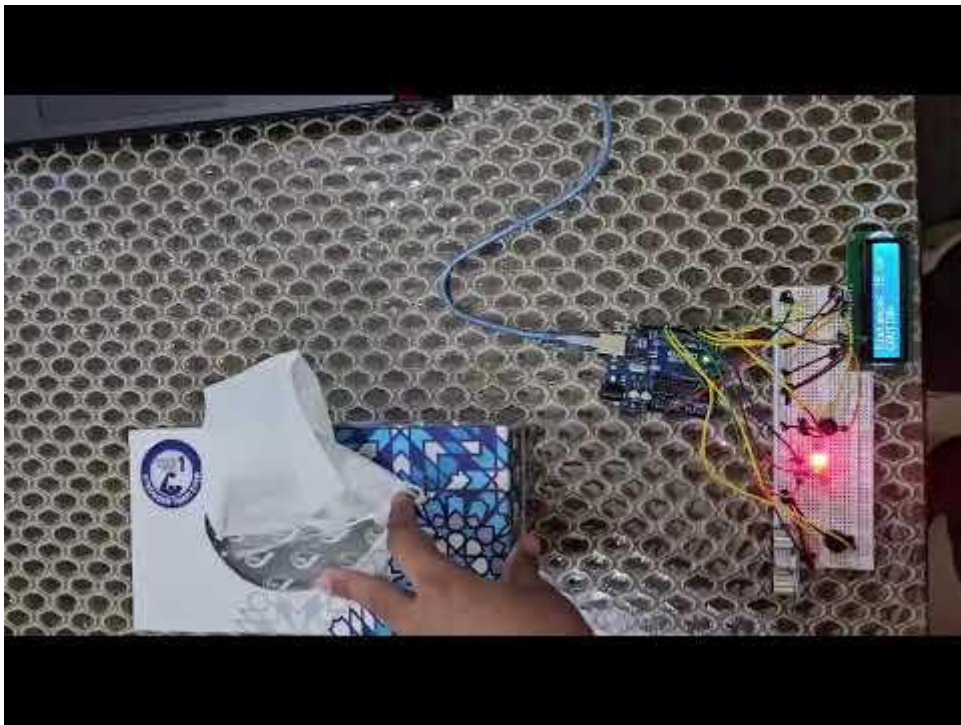
        tone(buzzer, 1000);
        delay(beepDelay);
        noTone(buzzer);
        delay(beepDelay);
    }
    else if (distance > 10) {
        lcd.setCursor(0, 1);
        lcd.print("CAUTION     ");
        digitalWrite(redLED, HIGH);

        tone(buzzer, 1000);
        delay(beepDelay);
    }
}

```

```
noTone(buzzer);  
delay(beepDelay);  
}  
else {  
  lcd.setCursor(0, 1);  
  lcd.print("STOP      ");  
  digitalWrite(greenLED, HIGH);  
  digitalWrite(yellowLED, HIGH);  
  digitalWrite(redLED, HIGH);  
  
  tone(buzzer, 1000);  
}  
}
```

## Video Demo





## Images

